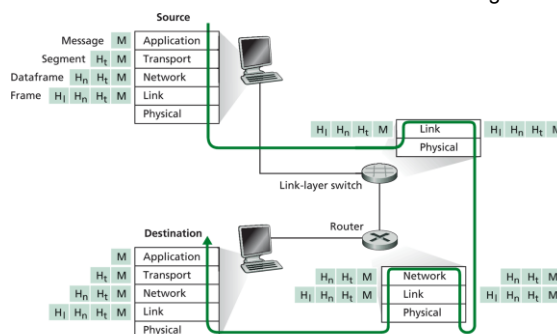


03 The Data Link Layer

03.01 Data link level design principles, Error control, Elementary data link protocols

Recall on layers: the encapsulation

- **Messages** from application level are passed to transport layer
- Transport layer concatenates (in the form of a header) additional information (that will be used by the receiver peer entity) building a **transport-layer segment**
- Data is passed to the network layer that adds its header and builds a **network-level datagram (or network-level packets)**
- Datagram is passed to the data link level that adds its header creating a **data link frame**



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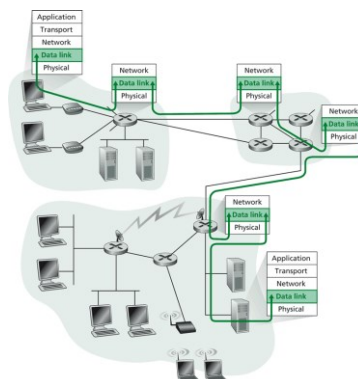
♦ Hosts, routers, and link-layer switches; each contains a different set of layers, reflecting their differences in functionality

Functions of the Data Link Layer

- Provide service interface to the network layer
- Dealing with transmission errors
- Regulating data flow
 - Slow receivers not swamped by fast senders

Functions of the Data Link Layer

- Data link layer's task is to provide a reliable and efficient communication between adjacent nodes (i.e. hosts/routers that are connected by a link)



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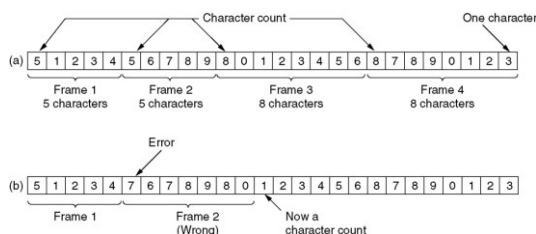
(a) • The link layer

Services offered by Data Link Layer

- Framing: framing structure is specified by the protocol
- Link access → Medium Access Protocol: very simple (or doesn't exist in point-to-point link)
 - When several nodes share a single channel MAC coordinates frames transmission
- Reliable delivery
 - Useful especially in links with high error rates (e.g. wireless links)
- Flow control
- Error detection (or correction)

Framing

- The bit stream that the physical layer supports is organized by the data link layer into frames. How can a bit flow divided in frames?
 - Character counting
 - Flag byte with byte stuffing
 - Flag with bit stuffing
 - Violations of physical layer encoding

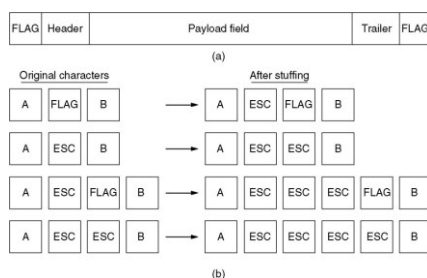


Since character count may, in case of errors, lose the synchronization, it is no longer used

A character stream. (a) Without errors. (b) With one error.

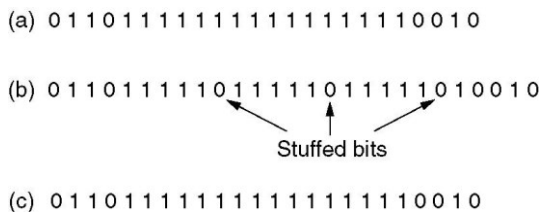
Framing

- Flag byte with byte stuffing: a flag byte delimits the beginning and the end of a frame
- What if flag byte is inside payload (e.g. in binary data is easy to happen)? → the source puts an escape character (ESC) before each occurrence
- What if ESC is in payload? → ESC again



Framing

- Character encoding may not be 8 bit (e.g. UNICODE chars are encoded as 16 bits)
- It is more flexible to allow arbitrary encoding → bit stuffing
- Each frame starts and ends with: 01111110
- The source inserts a '0' after each sequence of five '1'



Bit stuffing (a) The original data. (b) The data as they appear on the line. (c) The data as they are stored in receiver's memory after destuffing.

Framing

- Flag bytes with byte/character stuffing
 - Each frame start and end with special bytes – flag byte
 - If the flag byte occurs in the frame, stuff an extra escape byte (ESC).
 - Used in PPP (Point to Point Protocol)
- Starting and ending flags, with bit stuffing
 - Each frame begins and ends with a special bit pattern 01111110 whenever the sender sees 5 consecutive 1s, it stuffs a 0.
- Physical layer coding violations
 - Only applicable to networks in which the encoding on the physical medium contains some redundancy.
 - For example, a 1 bit is a high-low pair and a 0 bit is a low-high pair. High-high and low-low are used for delimiter.

Error Management

- A real Telecommunication system has transmission errors due to noise presence on the channel
- Many factors influences the error probability on sent bit
 - The average power of the received signal
 - The transmission speed
 - The noise density spectrum on the channel
 - The channel type (with additive or multiplicative noise, time variance characteristics , etc)

Error Management

- You can arbitrarily minimize the probability of error on the information bits by using appropriate channel encoding techniques.
- The channel encoding process consist of redundancy bit addition to the message that you want to transmit. At reception, the presence of such bits allows to detect or correct any errors introduced into the message by the noise on the channel.
- The disadvantage of channel encoding is that you have to transmit more bits of those that are actually needed to represent the message, on equal terms, increases the required time for transmission

Error Management

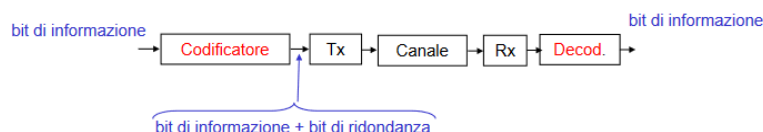
- Channel encoding, under the same conditions, does not change the probability of physically bits traveling on the channel.
- The use of channel encoding reduces the probability of error of information bits (or message bits).
- Below we will see the main coding strategies and the main codes that can be used to protect the message from the noise on the channel.

Strategies: Error Detection and Correction, Retransmission

- Error-detecting Codes
 - They are able to detect errors → source can be warned → re-transmission
- Error-correcting Codes
 - The destination is able to reconstruct the correct message from the corrupted one
- Both rely on the transmission of redundant information

Forward Error Correction Strategy

- (FEC): Use the codes to both detect errors and correct them (this is particularly useful in cases where retransmission of the message is not possible).



Codes: definitions

- Code Revelation Capacity: the ability to reveal the maximum number of errors in a code word.
- Code Correction Capacity: the maximum number of errors that the code correct in a code word.
- The performance of a code is measured based on
 - The ability to detect and / or correct errors
 - Efficiency $R_c = k / n$ ($k \rightarrow$ bit of message, $n \rightarrow$ length of the code word)
 - Complexity of execution

Error-control Codes

- Error detection and especially error correction codes are based on sophisticated mathematical techniques $\rightarrow$ modern algebra theory of fields
- We will only take into consideration 3 examples of error detection codes:
 - Parity check
 - Checksum method
 - Cyclic Redundancy Codes (CRC)

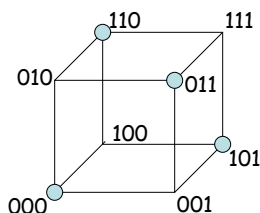
Hamming Distance

- In information theory, the **Hamming distance** between two strings of equal length is the number of positions at which the corresponding symbols are different.
- Put another way, it measures the minimum number of *substitutions* required to change one string into the other, or the number of *errors* that transformed one string into the other

Hamming distance for binary string

- For **binary strings** a and b the Hamming distance is equal to the number of ones in $a \text{ XOR } b$.
- The metric space of length- n binary strings, with the Hamming distance, is known as the *Hamming cube*; it is equivalent as a metric space to the set of distances between vertices in a hypercube graph.

Hamming cube



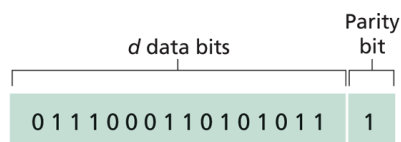
● codeword

- $d = 2 = \text{min distance}$
- $n = 3 = \text{dimensionality}$
- $2^n = 8 = \text{number of nodes}$

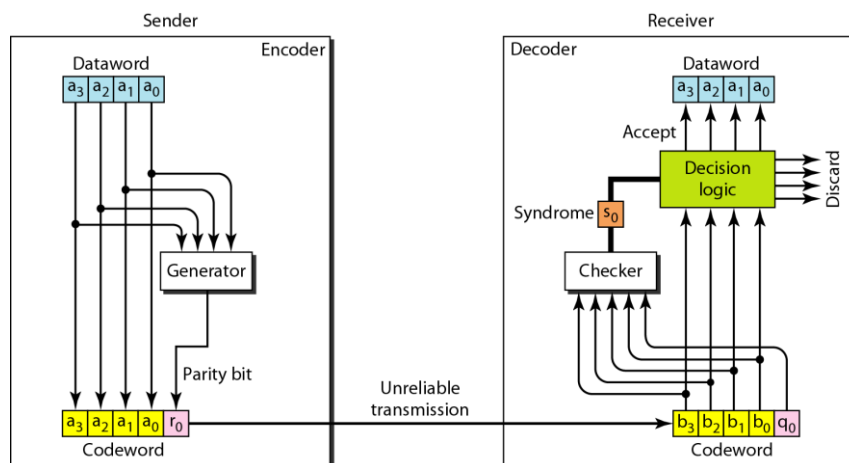
- The distance between nodes on the hypercube is the Hamming distance Δ . The minimum distance is d .
- 001 is equidistance from 000, 011 and 101.
- For s -bit error detection $d \geq s + 1$
- For s -bit error correction $d \geq 2s + 1$

Parity check

- Parity check consists in using an additional bit whose value is chosen to have an odd (odd parity) or even (even parity) number of '1'
- The receiver simply counts the number of '1' and checks if it is odd (even). If parity is not respected an error may have occurred
- If we have in even parity an even number of errors is not detected



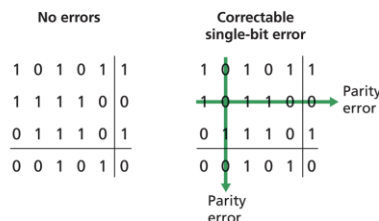
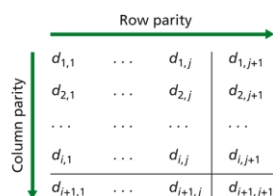
Parity check



- Encoder and Decoder for simply parity check code

Bi-dimensional parity check

- Bi-dimensional parity:
 - The d bits of data D are organized into i rows and j columns
 - For each row/column a parity bit is computed
 - It can be demonstrated that this technique can detect 2 errors combinations



Checksum

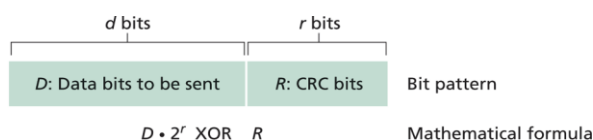
- The d bits of data are treated as if they were a sequence of k bit integers
- These integers are summed and complemented to 1: this is the checksum value transmitted together with data
- The receiver sums all the k bits integers: the result must be all '1'
- Used in Internet at transport level (low protection but easy software implementation)

Cyclic Codes

- The properties and advantage of this code are:
 - Their mathematical structure permits higher order correcting codes
 - Their structure can be easily implemented in hardware with shift registers and xor gates
 - Cyclic codes can be represented as polynomials
 - Cyclic code are all lateral (or cyclical)

Cyclic Redundancy Codes (CRC)

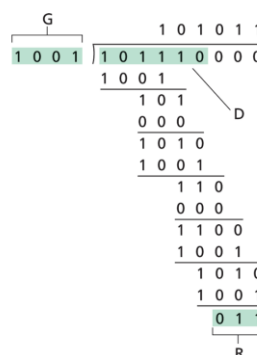
- The key concept is that strings of bits are seen as polynomials
- Transmitter and receiver agree on a common polynomial G (*generator polynomial*)
- Transmitter adds r bits to the d of data to obtain a $d+r$ binary string (T) that is divisible by G
- The receiver checks T/G : if remainder = 0 $\rightarrow$ ok, else $\rightarrow$ error

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♦ CRC codes

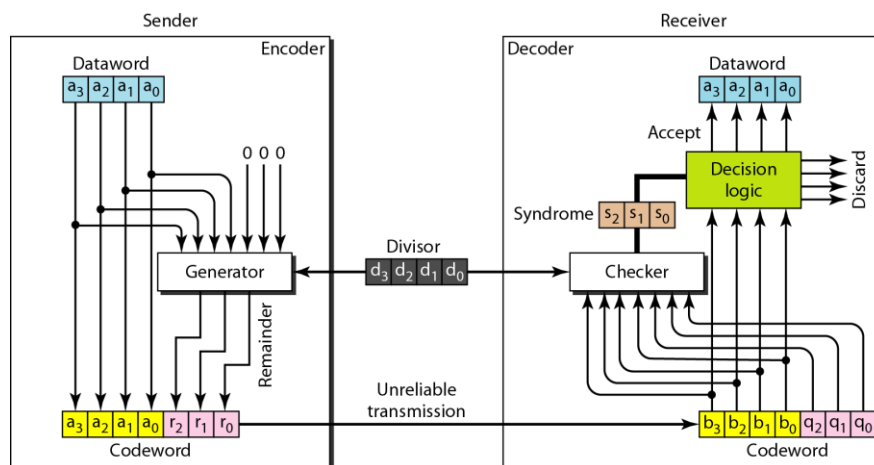
Cyclic Redundancy Codes (CRC)

- NB: division are polynomial divisions. Operations are mod 2 $\rightarrow$ + and - are equivalent to XOR
- How can R be computed?
- We want R : $D \cdot 2^r \text{ XOR } R = nG$
- $\rightarrow D \cdot 2^r = nG \text{ XOR } R$
- $\rightarrow R = \text{remainder of } (D \cdot 2^r / G)$
- Example: $D=101110$, $d=6$, $G=1001$, $r=3$
- Generator: standard definitions to have max detection probability with low redundancy
- CRC-32 (used in Ethernet):
 - $X^{32} + X^{26} + X^{23} + X^{22} + X^{16} + X^{12} + X^{11} + X^{10} + X^8 + X^7 + X^5 + X^4 + X^2 + X + 1$
- Others: CRC-8: $x^8 + x^2 + x^1 + 1$, CRC-10, CRC-12, etc.

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♦ A sample CRC calculation

Cyclic Redundancy Codes (CRC)



- CRC encoder and decoder

Cyclic Redundancy Codes (CRC)

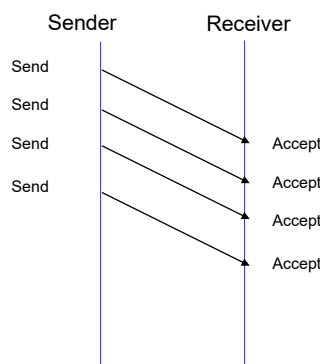
- If the generator has more than one term and the coefficient of x^0 is 1, all single errors can be caught.
- If a generator cannot divide $x^t + 1$ (t between 0 and $n - 1$), then all isolated double errors can be detected.
- A generator that contains a factor of $x + 1$ can detect all odd-numbered errors.
- A good polynomial generator needs to have the following characteristics:
 - It should have at least two terms.
 - The coefficient of the term x^0 should be 1.
 - It should not divide $x^t + 1$, for t between 2 and $n - 1$.
 - It should have the factor $x + 1$.

Elementary Data Link Protocols

- Reliable data link transmission is achieved through:
 - ACK
 - Timeout
 - ACK+timeout = ARQ (Automatic Repeat Request Protocol)
- We will analyze some examples of increasing complexity:
 - An Unrestricted Simplex Protocol
 - A Simplex Stop-and-Wait Protocol
 - A Simplex Protocol for a Noisy Channel (1st ARQ example)
 - Sliding window protocols

A very simple protocol: unrestricted and simplex

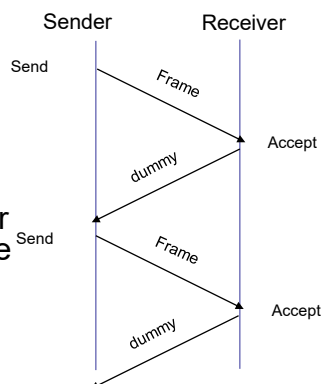
- Assumptions:
 - Data is transmitted in one direction (simplex channel) on an error free channel
 - Physical and network layers always ready
 - Infinite time and buffer space available
 - Communication channel never loses frames



Stop and wait protocol

Assumptions:

- Data transmitted in one direction
- Physical and network layers not always ready
- Error free channel
- The solution is that receiver sends back a dummy frame that allows the transmitter to send a new frame



Stop-and-Wait Protocol for noisy channels

•What if we consider the possibility of errors? We introduce this protocol solution (ARQ) based on ACK+timeout

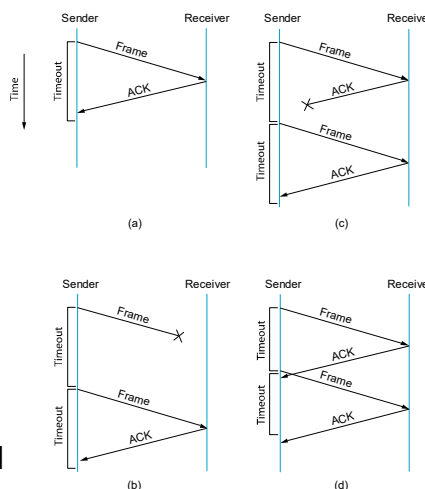
•Look at the timing diagrams in 4 different cases for stop and wait algorithm

•(a) ACK frame received before timeout

•(b) original frame is corrupted

•(c) ACK frame is corrupted

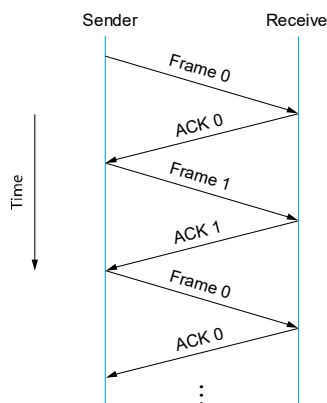
•(d) early timeout



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Stop-and-Wait Protocol for noisy channels

- Sequence numbers need to be introduced since in case an ACK is lost or timeout is declared too early, the receiver could receive 2 copies of the same frame
- Timing diagram for stop and wait algorithm with 1 bit sequence numbers
- Characteristics: only 1 frame at the time is on the line → bandwidth is not optimized



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Something to think about..

A taxonomy of error correction

ARQ			FECC	
Stop & wait	Continuous ARQ (pipelining)		Block codes	
	Go-back-N	Selective repeat	Others (non-linear)	Group (linear)
			Others (non-cyclic)	Polynomially generated (cyclic)

Automatic Repeat and ReQuest strategies

- Automatic Repeat and ReQuest (ARQ): Use the codes only to detect the presence of errors in the received message. When a fault is detected, the receiver asks for the retransmission of the message to the transmitter (it needs a reverse channel).

